Project Documentation

Defect Tracking & Management System

Domain: Software Project Management
Topic: Basic Bug/Issue Tracker

Submitted by:
Jeeten Misra
Roll No: 23052397
Program: B. Tech CSE
Batch: SAP ABAP 2027
Institution: Kalinga Institute of Industrial Technology (KIIT)
Bhubaneswar, Odisha

1. Executive Summary

The Defect Tracking & Management System (DTMS) brings everything into one web-based platform, managing the entire lifecycle of a defect from the moment it's reported to its final resolution. Along the way, it reflects key Software Project Management principles, such as clearly defined workflow state transitions and structured task assignment, helping teams stay organized, accountable, and efficient.

Core Features

- End-to-End Workflow: Enforces strict state transitions (Open -> In Progress -> Resolved -> Closed).
- Kanban Board: A visual dashboard to map the lifecycle of a defect in real-time.
- Issue Reporting: Capture detailed bug reports with priority tags (Low, High, Critical).
- Task Triaging: Direct assignment of tickets to specific developers via an intuitive interface.

2. Theoretical Framework & Tech Stack

The SAP SD Order-to-Cash (O2C) Connection

The Order-to-Cash (O2C) cycle in SAP Sales & Distribution represents a complete, end-to-end sales journey, moving through clearly defined stages like Quotation, Sales Order, Delivery, and finally Billing. Each step follows a strict document flow, where transitions are controlled to maintain accuracy and process integrity. The Defect Tracking & Management System is built on this same kind of enterprise-grade logic.

In SAP, for example, a user can't generate an invoice unless the delivery step has already been completed. (DTMS) applies that same discipline to issue tracking. It enforces structured state validation, ensuring that a developer can't simply close a bug at will it must first move through active development and pass Quality Assurance verification. This approach keeps the workflow consistent, reliable, and aligned with real-world process standards.

Technology Stack

- Frontend: React.js (Functional components, Hooks) for the interactive Kanban interface.
- Backend: Node.js with Express to handle API routing and state updates.

Database: Lightweight SQLite / JSON-based structure for rapid prototype deployment.

3. End-to-End Workflow Implementation

Step 1: Bug Logging (Status: Open)

When a Quality Assurance tester runs into an error, they log it as a new issue here it is a “TestBug” and mark it with a high priority so it gets immediate attention. The system then places it on the Kanban board under the “Open” column, where it sits as unassigned until a team member picks it up.

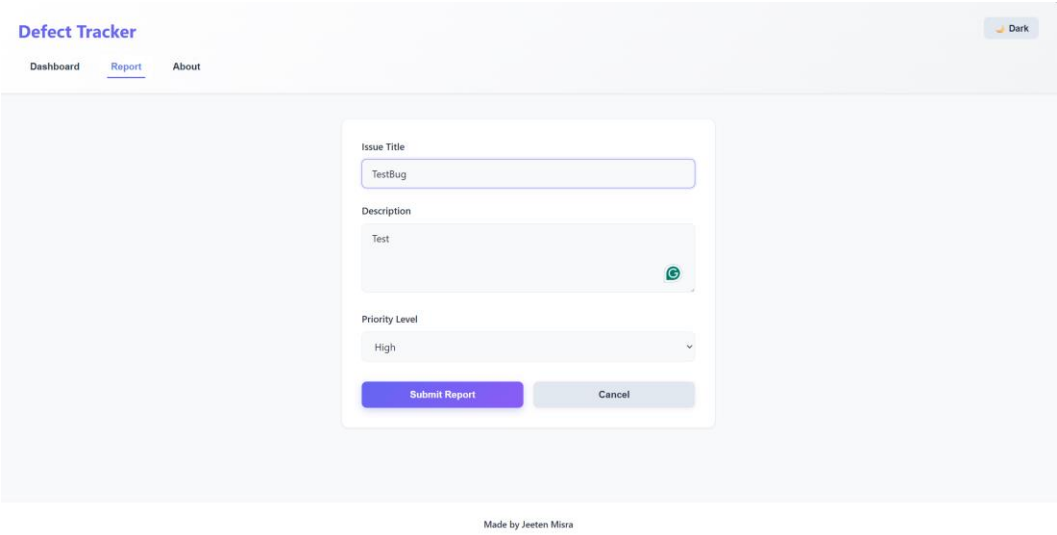

Figure 1: Issue Reporting Module

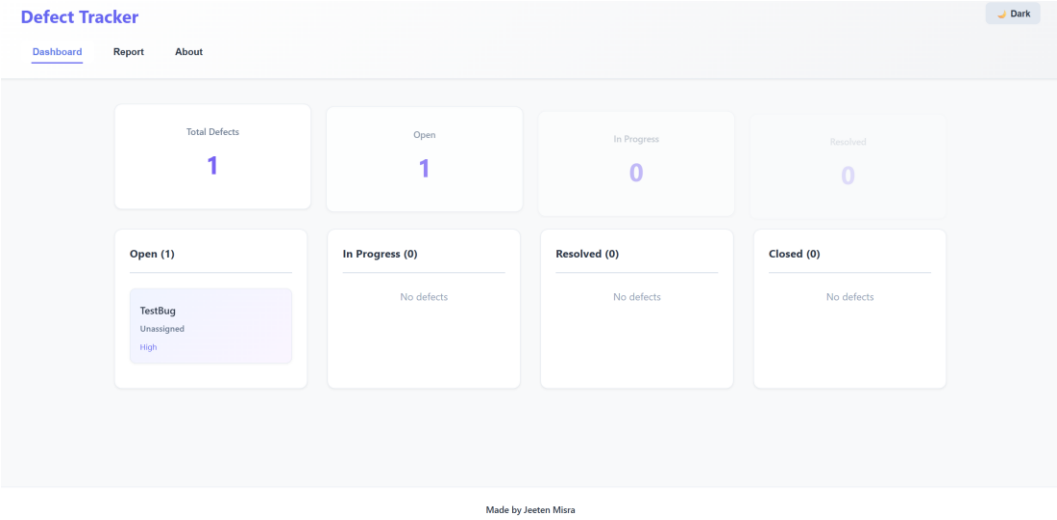

Figure 2: Dashboard Registration (Open State)

Step 2: Triaging and Assignment

A Project Manager opens the Edit model, updates the status to "In Progress", and assigns it to a developer (Jeeten).

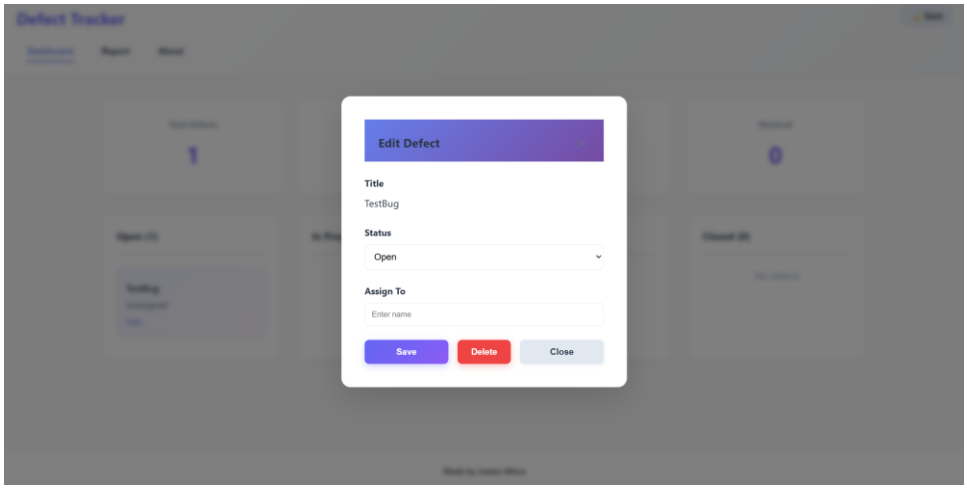

Figure 3: Edit Modal for Assignment

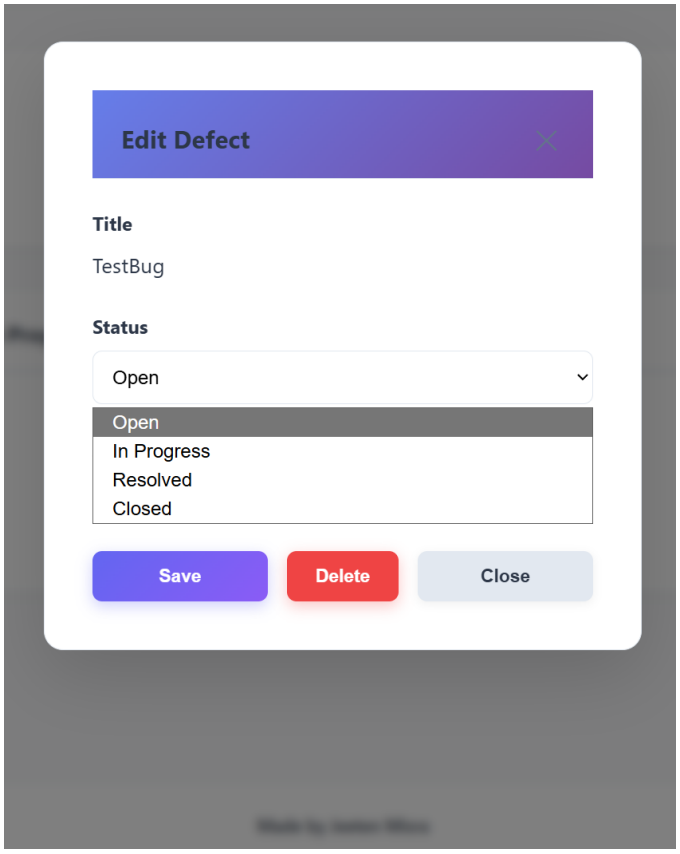

Figure 4: Status Change Dropdown

Step 3: Active Development & Final Closure

As work begins, the defect moves into “In Progress” and later shifts to “Resolved” once a fix has been applied. After that, a Quality Assurance tester steps in to verify the solution. If everything checks out, they mark the issue as “Closed,” effectively wrapping it up and archiving the task.

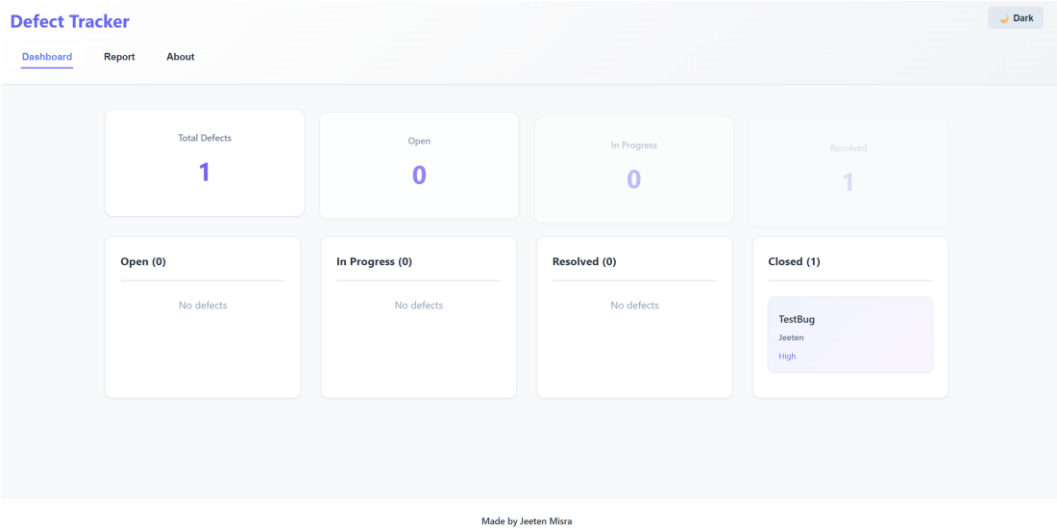

Figure 5: Final State of the Workflow (Closed)

4. Conclusion & Future Improvements

The Defect Tracking & Management System (DTMS) turns complex ERP-style state transition logic into something much lighter and easier to use in day-to-day project management. By narrowing its focus to just the essential tracking needs, it stays fast, simple, and easy for teams to adopt without friction.

Looking ahead, there are several practical ways to make it even stronger. Adding Role-Based Access Control would help ensure that only the right people like Admins, Developers, and Quality Assurance can move tasks between certain states, improving both security and accountability. Automated notifications could keep everyone in the loop by sending email alerts whenever an issue is escalated or updated. Integrating version control systems like GitHub would also be a big win, allowing code commits to automatically update or transition related Kanban tickets, reducing manual work and keeping everything in sync.